15 Continuous delivery and GitOps

This chapter covers

- Understanding continuous delivery and release management
- Configuring Spring Boot for production with Kustomize
- Deploying in production with GitOps and Kubernetes

Chapter after chapter, we have gone through patterns, principles, and best practices for working with cloud native applications, and we've built a bookshop system using Spring Boot and Kubernetes. It's time to deploy Polar Bookshop to production.

I expect you have the projects of the Polar Bookshop system in separate Git repositories stored on GitHub. If you haven't followed along in the previous chapters, you can refer to the Chapter15/15-begin folder in the source code accompanying the book, and use it as a foundation to define those repositories.

This chapter will guide you through some final aspects of preparing applications for production. First I'll discuss versioning strategies for release candidates and how to design the acceptance stage of a deployment pipeline. Then you'll see how to configure Spring Boot applications for production and deploy them on a Kubernetes cluster in a public cloud. Next, I'll show you how to complete the deployment pipeline by implementing the production stage. Finally, you'll use Argo CD to implement continuous deployment based on the GitOps principles.